SAMARTH

CUET PG EXAM





Lecture - 2 | Computer Science

Digital Fundamental

Topic : Number System and Conversion Part-9



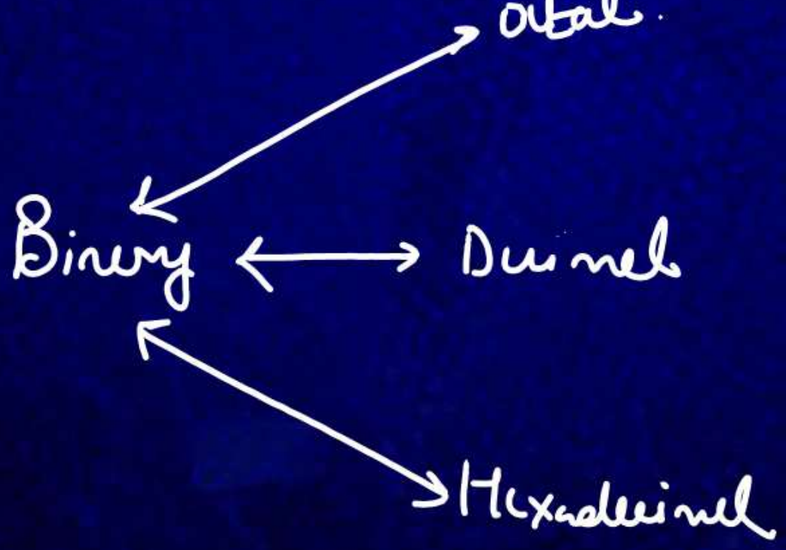
By- Puneet Kaur

Topics *to be covered*

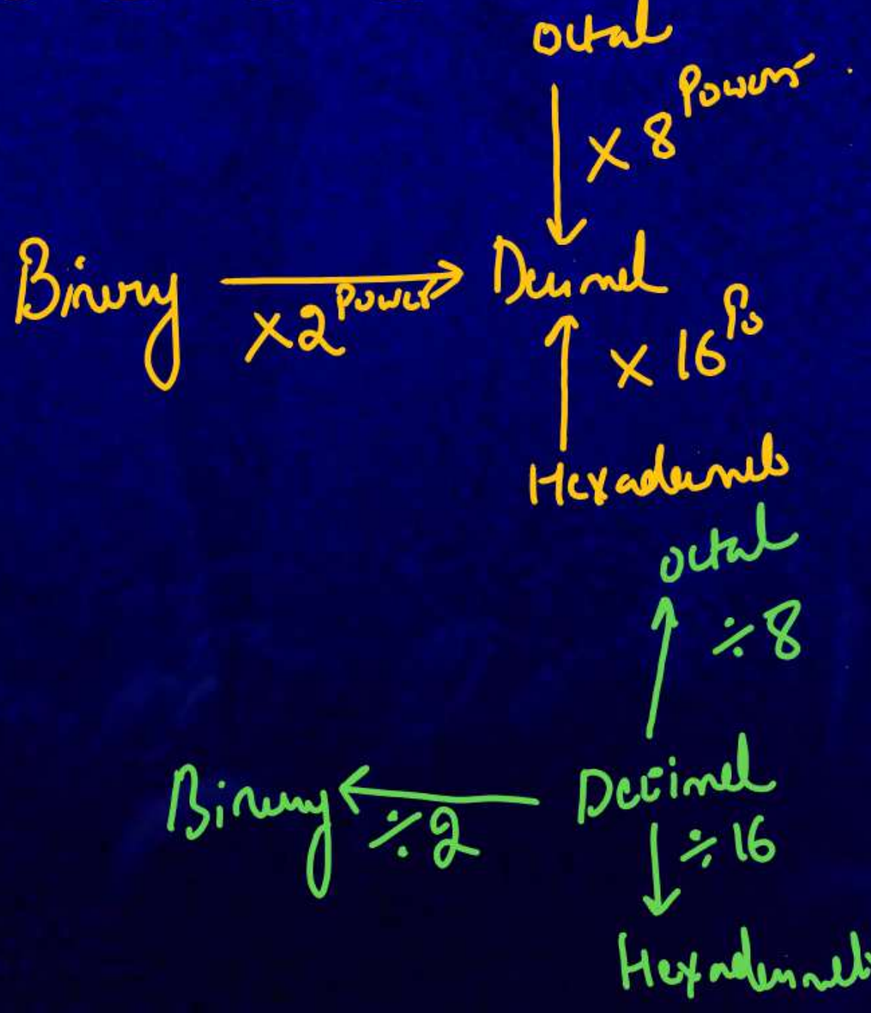
1. Number System and Conversion Part-1



Topic : Number System and Conversion Part-1



Trick to Remember :



(7) Octal to Decimal

$(\ )_8$ to $(\ )_{10}$

Ex. $(724)_8$

7 2 4

- $4 \times 8^0 = 4$
- $2 \times 8^1 = 16$
- $7 \times 8^2 = 448$

$+$

468

Octal

000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

(9) $(\text{Decimal})_{10}$ to $(\text{octal})_8$

Ex. $(914)_{10} = (?)_8$

8 | 914	
8 | 114	2
8 | 14	2
	1 6

$(1622)_8$

(8) Hex. to Decimal.

$()_{16}$ to $()_{10}$

13

8 2 D

D → $\times 16^0 = 13$

2 → $\times 16^1 = 32$

8 → $\times 16^2 = 2048$

\+

2093.

Hex

0	0	0	0	— 0	
0	0	0	1	— 1	
0	0	1	0	— 2	
0	0	1	1	— 3	
0	1	0	0	— 4	
0	1	0	1	— 5	
0	1	1	0	— 6	
0	1	1	1	— 7	
1	0	0	0	— 8	
1	0	0	1	— 9	
1	0	1	0	— 10	A
1	0	1	1	— 11	B
1	1	0	0	— 12	C
1	1	0	1	— 13	D
1	1	1	0	— 14	E
1	1	1	1	— 15	F

(10) Decimal to Hexadecimal.

$()_{10}$ to $()_{16}$

Ex

$(90)_{10} = (?)_{16}$

16	80
	5 0

(50)

16	90	A
	5	(10)

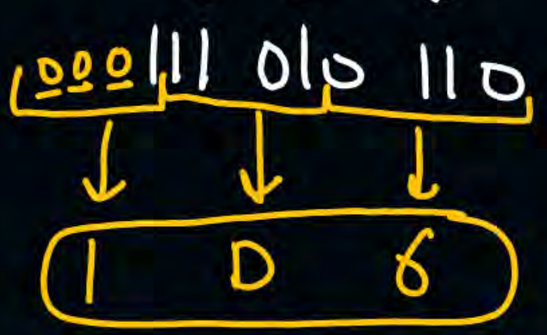
5A

(11) Octal to Hexadecimal.

$(\ )_8$ to $(\ )_{16}$

① Octal to Binary.
② Binary to Hex

Ex. $(726)_8 = (?)_{16}$.

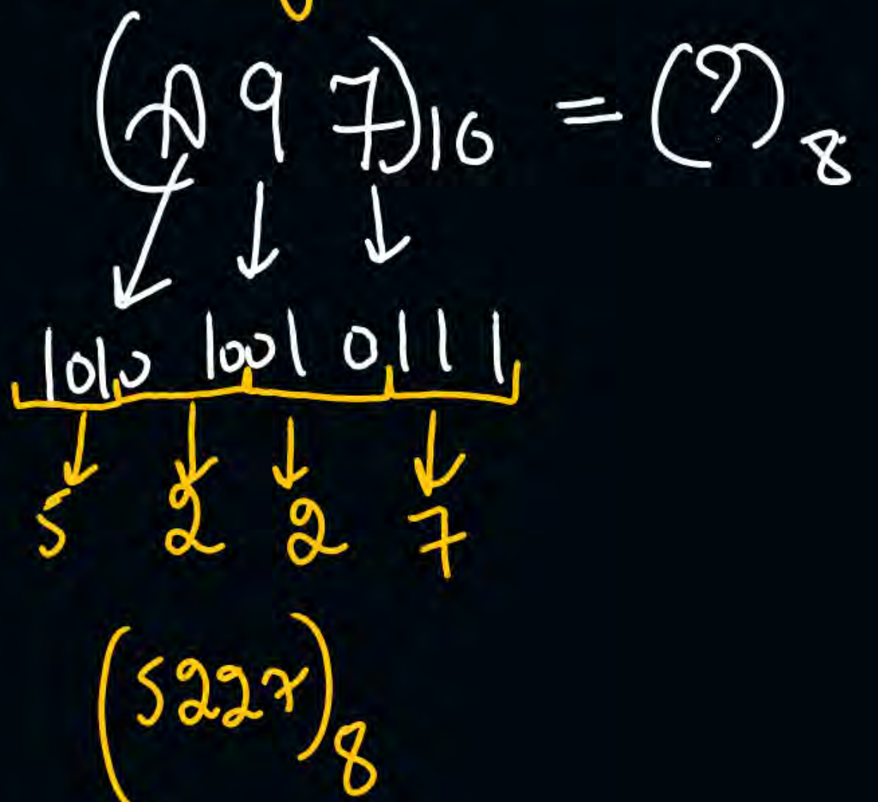


(12) Hex to Octal.

$(\ )_{16}$ to $(\ )_8$

① Hex to Binary
② Binary to Octal

Ex. $(A97)_{16} = (?)_8$



$(5227)_8$

Topic : Number System and Conversion Part-1

Bit	One binary digit
Byte	8 bits
Kilobit	1,024 or 2^{10} bits
Kilobyte	1,024 or 2^{10} bytes
Megabit	1,048,576 or 2^{20} bits
Megabyte	1,048,576 or 2^{20} bytes
Gigabit	2^{30} bits
Gigabyte	2^{30} bytes
Terabyte	2^{40} bytes
Petabyte	2^{50} bytes
Exabyte	2^{60} bytes

Topic : Number System and Conversion Part-1

Binary, octal, decimal, and hexadecimal are all number systems used in digital electronics and modern computing:

Decimal

- Uses base 10 and the place values are based on the powers of 10. Any number can be represented using the digits 0–9.

Binary

- Uses base 2 and each digit can only be a 0 or 1.

Octal

- Uses base 8 and the value of each place is based on the powers of 8. Any number can be represented using the digits 0–7.

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Hexadecimal

- Uses base 16 and the value of each place is based on the powers of 16. Any number can be represented using the digits 0–9 and the characters A, B, C, D, E, and F.
- To convert between bases other than decimal, it's usually necessary to first convert the number to base 10, and then convert that number to the second base. However, it's easy to convert directly between binary and octal, and between binary and hexadecimal.

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Here are some ways to convert between number systems:

- Decimal to binary: Divide the number by 2.
- Decimal to hexadecimal: Divide the number by 16.
- Decimal to octal: Divide the number by 8 and write the remainders in reverse order.

Topic : Number System and Conversion Part-1

#Q. Convert the binary number $(1101)_2$ into a decimal number.

A. 12

B. 13

C. 14

D. 15

Topic : Number System and Conversion Part-1

Ans. B

Topic : Number System and Conversion Part-1

#Q. Convert 0.1011 to decimal.

A. 0.6875

B. 0.6857

C. 0.5867

D. 0.7865

Topic : Number System and Conversion Part-1

Ans. A

Topic : Number System and Conversion Part-1

#Q. Convert decimal number 27 to binary.

A. 110011

B. 11001

C. 11011

D. 11101

Topic : Number System and Conversion Part-1

Ans. C

Topic : Number System and Conversion Part-1

#Q. Convert decimal number 10.25 to binary.

A. 1010.10

B. 1010.01

C. 1010.00

D. 1010.11

Topic : Number System and Conversion Part-1

Ans. B

Solution:

$(10.25)_{10} = (?)_2$

For 10:

2	10
2	5 --- 0
2	2 ---1
2	1 --- 0
2	0 --- 1

Radix Point

= 1010

For .25:

2 * 0.25 = 0.50

2 * 0.50 = 1.00

= 01

Result is: $(1010.01)_2$

Topic : Number System and Conversion Part-1

#Q. Convert Binary number 11101011 to octal.

A. 352

B. 253

C. 353

D. 453

Topic : Number System and Conversion Part-1

Ans. C

Topic : Number System and Conversion Part-1

#Q. Convert binary number 101010011.110100 to octal.

A. 523.64

B. 533.64

C. 543.65

D. 533.56

Topic : Number System and Conversion Part-1

Ans. A

Topic : Number System and Conversion Part-1

#Q. Convert decimal number 540 to octal.

A. 1033

B. 1035

C. 1034

D. 1036

$(\quad)_{10}$ to $(\quad)_8$

8 | 540

Topic : Number System and Conversion Part-1

Ans. C

Decimal to Octal

8	540	
8	67	4
8	8	3
8	1	0
	0	1

$(540)_{10} = (1034)_8$

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#Q. Convert Decimal number 29.30 to octal.

A. 35.251

B. 35.231

C. 35.341

D. 35.355

$\cdot 30 \times 8 =$

Ex. $(42.50)_{10} = (?)_2$

$\cdot 50 \times 2 = 1.00$

$\cdot 00 \times 2 = 0$

$(101010.10)_2$

2	42	
2	21	0
2	10	1
2	5	0
2	2	1
	1	0

Topic : Number System and Conversion Part-1

Ans. B

8	29

29.30

8	29	
8	3	5
	0	3

$(29.30)_{10} = (35.23146)_8$

$0.30 \times 8 = 2.4$
$0.40 \times 8 = 3.2$
$0.20 \times 8 = 1.6$
$0.60 \times 8 = 4.8$
$0.80 \times 8 = 6.4$
$0.40 \times 8 = 3.2$ — Stop
$0.20 \times 8 = 1.6$ — Repeated
$0.60 \times 8 = 4.8$
$0.80 \times 8 = 6.4$

Q. Octal no

326.4 is equivalent to

(1) $(214.2)_{10}$ & $(D6.8)_{16}$

(2) $(212.5)_{10}$ & $(1D6.8)_{16}$

(3) $(214.5)_{10}$ & $(D6.8)_{16}$

(4) $(214.5)_{10}$ & $(D6.4)_{16}$

$326.4 \longrightarrow (\text{Octal})_8$

$\times 8^{-1}$

$\times 8^0 = 6$

$\times 8^1 = 16$

$\times 8^2 = 192$ +

214.5

$8^{-1} = \frac{1}{8} \times \frac{4}{4} = \frac{2}{2^3} = \frac{1}{2^2} = 0.5$

① Octal to Binary

② Binary to Hexadecimal

Ex: 326.4

011 010 110 . 100

0 0 0 0 1 1 0 1 0 1 1 0 . 1 0 0 0

0 D 6 8

4 = 0100

8_2 1000

2 Mins Summary

Number System

THANK YOU